

RICHARDSON ALMEIDA

Backend Software Engineer | Python & Flask | AI-Integrated Systems

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Summary

Backend Software Engineer with 10+ years in technology, including 4+ years applying Python to production systems on national-scale public platforms. Hands-on experience with Python, Flask, SQLAlchemy, PostgreSQL, REST APIs, Docker, and CI/CD on Linux environments. Currently pursuing an MBA in Artificial Intelligence for Business, with active work on AI-integrated systems (LLM, OpenAI/Whisper) and formal testing practices. Open to backend engineering roles (Python, Flask, REST APIs), remote, internationally.

Experience

Backend Software Engineer (Development Coordinator & Technical Leadership)

MINISTRY OF AGRICULTURE (MAPA) | Brasília, Brazil (on site)

Feb 2023 - Aug 2026

Coordinated backend engineering and systems integration across national-scale public digital platforms, working with development teams, architects, and institutional stakeholders. Applied Python, Flask, PostgreSQL, Docker, and REST APIs to guide technical decisions and validate solutions. Core areas: systems integration, legacy modernization, infrastructure, and data-privacy governance.

- Prototyped a lightweight intake solution (Google Forms) to quickly validate a structured request-tracking approach, applying Python/Flask/PostgreSQL-oriented engineering judgment to the problem; the resulting case justified the organization's adoption of a full ticketing system (OTRS), moving from 0% to 100% traceability of request origin, timing, and status.
- Coordinated internalization of SISPA (Java, Spring Boot, Angular, Oracle), integrating MAPA, ANVISA, and IBAMA via REST APIs to unify previously manual and fragmented registration and evaluation workflows.
- Coordinated modernization of SHIVA-Viajantes (Node.js, React, PostgreSQL, AWS), replacing manual, form-based tracking across 90+ embarkation points nationwide (airports and border checkpoints) with a structured, risk-based inspection platform for MAPA's VIGIAGRO program.
- Initiated and structured the modernization of the SDA Inspection System (Nest.js, React Native, Redis, PostgreSQL), defining the technical foundation to replace paper-based workflows with mobile field tooling; handed off the project ready for continued development following departure from the Ministry.
- Took over VITRINE 2 (Next.js, TypeScript, PostgreSQL) before final MVP scope was closed, directing stabilization and REST integrations with federal registries (CAF, Bcadastro), and leading the retirement of the legacy system it replaced. The platform is now in production and under continued incremental development.
- Proposed a screen-prototyping approach for EXPOV before a coordinator was formally assigned, validating the solution early and enabling on-time delivery; later structured its integration (PostgreSQL, Oracle) with SIPEAGRO export quality data.

Skills: Python, SQL, Flask, SQLAlchemy, REST APIs, PostgreSQL, MySQL, Docker, CI/CD, Linux, Git, GitHub, OpenAI API, Whisper, Backend Architecture, Systems Integration, Legacy Modernization, Modular Design.

Technical Coordinator (Automation & Administrative Systems)
INSTITUTO FEDERAL DE BRASÍLIA (IFB) | Brasília, Brazil (on site)
Feb 2010 – Feb 2023 · 13 yrs 1 mo · Brasília, Brazil · On-site

Coordinated technical and administrative operations across a federal education institution for 13 years. From 2015 onward, shifted focus to process automation, systems structuring, and operational efficiency using Python, SQLite, and Google Workspace tools.

- Developed Python scripts to automate student record structuring and label generation, replacing manual processes and enabling consistent file lookup by name, enrollment number, or CPF across the institution's administrative workflow (Python, SQLite).
- Built a web-based intake form (Google Forms) to register student requests and control issuance of academic statements, bridging an operational gap while the official academic records system was not yet in place (Google Forms, Google Sheets).
- Automated CPF cross-checking to prevent duplicate registrations and auto-populated folder labels from a Google Sheets data source (Python, SQLite, Google Sheets).
- Supported the first institutional adoption of a flexible working-hour policy (40h → 30h weekly) in collaboration with the career committee, union representatives, and the rector's office.

Skills: Python, SQLite, Google Forms, Google Sheets, Google Workspace, Process Automation, Workflow Automation, Scripting, Data Structuring, Data Validation, Data Integration, Web Form Development, Systems Structuring, Operational Efficiency, Legacy Process Replacement, Cross-functional Coordination, Stakeholder Alignment, Institutional Policy Support, Operational Analysis

Education

MBA in Artificial Intelligence for Business – Faculdade Exame in partnership with Saint Paul Escola de Negócios, Brazil - 2026 (in progress)

Bachelor's Degree in Systems Analysis and Development – Universidade Paulista, Brazil (2016 - 2020)

Technical Degree in Systems Development – Instituto Federal de Brasília (IFB), Brazil (2018 - 2021)

Postgraduate Certificate in Professional and Technological Education – Instituto Federal de Goiás (IFG), Brazil (2018 - 2021)

Languages

English: Full professional proficiency

Portuguese: Native